

Jason Downing

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Technical Skills

Languages & Databases: Python, Go, JavaScript, C#, SQL (PostgreSQL, SQLite, MySQL, SQL Server)

Frameworks & Web: Django, REST APIs, .NET / ASP.NET, Microservices, HTML5/CSS3

Tools & DevOps: Docker, Linux (RedHat), Git, GitHub, Jenkins, CI/CD Pipelines, InfluxDB, Grafana, Vagrant

Employment

Software Engineer | ReversingLabs

10/2021 - Present

- Architected and redesigned enterprise High Availability (HA) and redundancy systems on RedHat/Rocky Linux for core threat analysis platforms (Spectra Analyze & Detect Manager), satisfying key enterprise reliability requirements.
- Maintained and scaled high-throughput malware hunting platforms (Spectra Analyze & Detect), interfacing REST/Python backends with complex binary analysis engines across Kubernetes microservices and virtual appliances (OVAs).
- Architected REST API pipelines and integrations for third-party sandbox and analysis solutions (Cisco SMA, VMRay, Auxiliary Analysis), enabling automated file ingestion, analysis, and report aggregation.
- Served as a core member of the escalation engineering team, driving rapid resolution for high-priority enterprise customer demands, complex bug fixes, and critical feature enhancements.
- Mentored and onboarded junior engineers, co-ops, and summer interns; conducted code reviews, provided technical guidance, and accelerated time-to-productivity for new team hires.
- Engineered backend configuration services for Titanium Scale worker/hub deployments using CentOS, Vagrant, Docker, and VirtualBox within CI/CD pipelines (Jenkins, Git).

Software Engineer II | Raytheon Technologies

08/2019 - 10/2021

- Architected Six Sigma automation solution integrating Python REST APIs with InfluxDB and Grafana dashboards, delivering real-time metrics and saving over \$50,000 annually.
- Engineered Java-based microservices using Docker containers, Maven, Git, and Jenkins in an Agile/Scrum environment.
- Designed Python simulation system using COTS products to emulate binary data bus traffic, enabling comprehensive integration testing for embedded systems.
- Overhauled legacy Python test codebase into modular, object-oriented design and built a custom GUI to streamline hardware configuration testing.

System Engineer II | BAE Systems

04/2018 - 08/2019

- Developed full-stack web application with C#, ASP.NET, JavaScript, and SQL Server replacing complex manual spreadsheet workflows, resulting in \$100,000+ annual operational savings.
- Created custom Python security-auditing tools to scan repository codebases for disallowed terms and compliance violations.
- Enhanced C# .NET automation system with full-stack interfaces using AngularJS, Bootstrap, jQuery, and SQLite.

Software Developer | Creative Logistics Solutions

07/2017 - 04/2018

- Architected XML REST API web integration for international express delivery carrier, successfully replacing a legacy licensed third-party component.
- Implemented C# .NET web service for regional overnight shipping carrier, delivering end-to-end functionality one week ahead of schedule.
- Integrated government REST API for export compliance screening, enabling automated validation of regulated trade partners.

Education

University of Massachusetts Lowell

Bachelor of Science in Computer Science